

PhysLive E2E Physics Verification Report

Three teacher prompts were exercised through the backend extraction, ambiguity confirmation, schema routing, solver, and validation pipeline.

Execution time: 2026-09-23 00:48 SE Asia Standard Time

Case	Schema	Result	Validation
1. Uniform acceleration	kinematics@1.11	PASS	passed
2. Projectile at 45 deg	kinematics_projectile@1.11	PASS	passed
3. Two-body elastic collision	dynamics_collision@2.0	PASS	passed

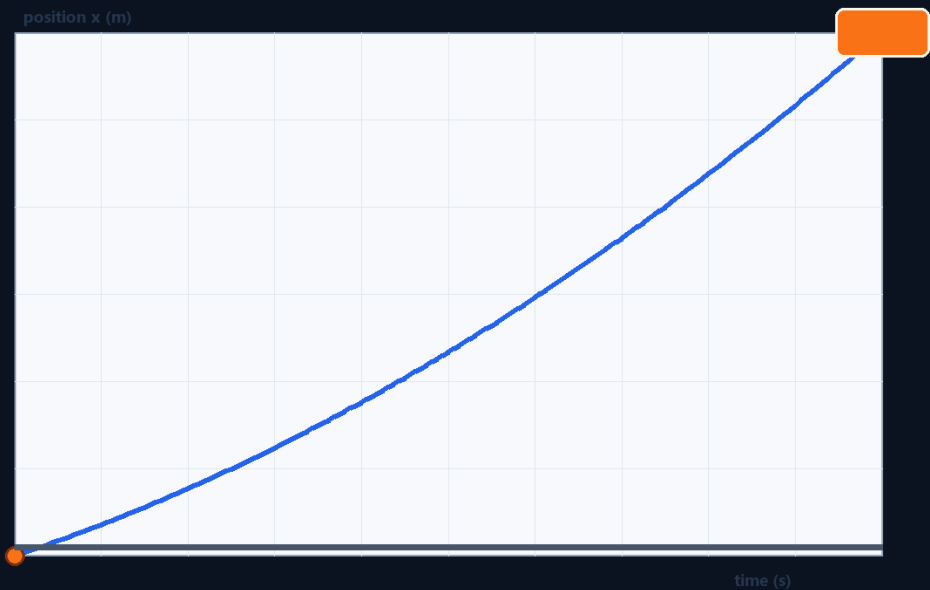
Test account: teacher@gmail.com (password omitted from this report). API: localhost:8080. Frontend static architecture, scene contracts, lint, 30 tests, and production build also passed.

Case 1 - Uniform acceleration

Prompt: Một vật chuyển động thẳng với vận tốc đầu 10 m/s và gia tốc 2 m/s². Hãy mô phỏng trong 8 giây.

Case 1 - Uniformly accelerated motion

Backend-validated scene snapshot | schema kinematics@1.11



PASS

t = 8.00 s

x = 144.00 m

v = 26.00 m/s

a = 2.00 m/s²

Initial conditions

v0 = 10.00 m/s

x0 = 0.00 m (confirmed)

Validation: passed

No unresolved ambiguity

Confirmed missing input	initial_position = 0 m
Resolved end	time_limit at 8.00 s
Output	x = 144.00 m, v = 26.00 m/s, a = 2.00 m/s ²
Simulation / run	0a6fd479-73c8-481c-93bf-d8aa5aefd958 / 5175600e-5bf3-41c1-a8bb-9a6736ad0753

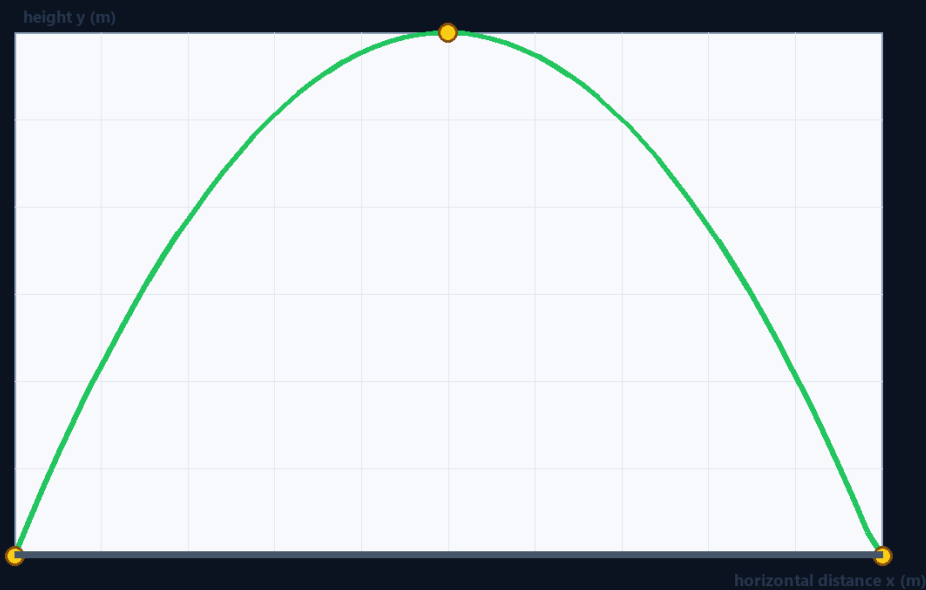
Validation result: PASS (all analytical checkpoints passed within tolerance 0.02).

Case 2 - Projectile motion

Prompt: Ném một vật với vận tốc đầu 20 m/s, góc ném 45°, bỏ qua sức cản không khí.

Case 2 - Projectile motion

Backend-validated scene snapshot | schema kinematics_projectile@1.11



PASS

$v_0 = 20.00 \text{ m/s}$
 $\text{angle} = 45 \text{ deg}$
 $g = 9.81 \text{ m/s}^2$
 $\text{landing } x \approx 41.01 \text{ m}$

Confirmed inputs

initial position = 0 m
initial height = 0 m

Validation: passed

Rendered until ground contact

Confirmed missing inputs	initial_position = 0 m; initial_height = 0 m
Resolved end	catalog default time_limit at 10.00 s; snapshot clipped at ground contact
Output	$v_0 = 20.00 \text{ m/s}$, $\text{angle} = 45 \text{ deg}$, $g = 9.81 \text{ m/s}^2$
Simulation / run	4e14e079-fe6b-4aaa-9565-92516f54bc77 / f4939048-f2fc-4bfe-b012-8190995a75eb

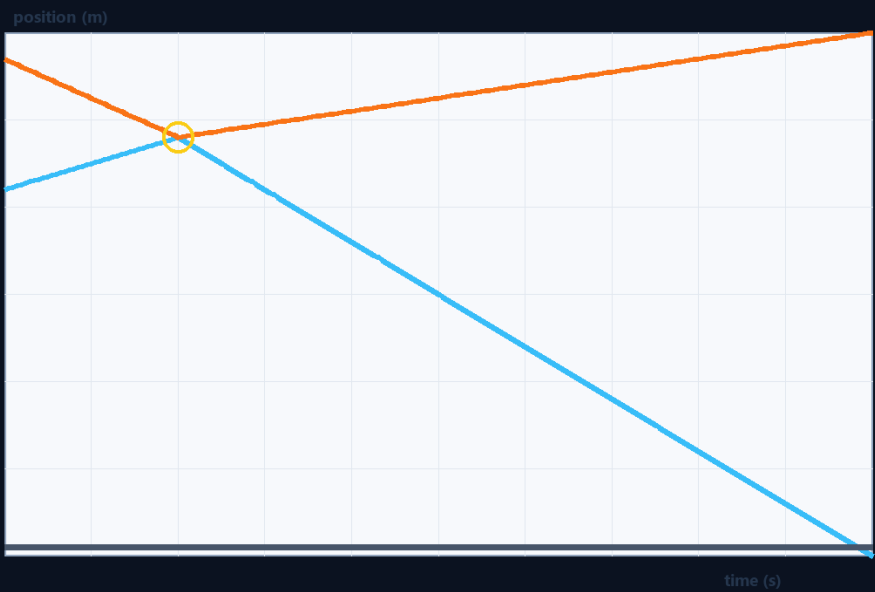
Validation result: PASS (all analytical checkpoints passed within tolerance 0.02).

Case 3 - Elastic collision

Prompt: Hai vật có khối lượng 2 kg và 3 kg chuyển động ngược chiều rồi va chạm đàn hồi.

Case 3 - Elastic collision

Backend-validated scene snapshot | schema dynamics_collision@2.0



PASS

m1 = 2.00 kg
m2 = 3.00 kg
v1 before = +2.00 m/s
v2 before = -3.00 m/s
v1 after = -4.00 m/s
v2 after = +1.00 m/s

Collision event resolved
t ~ 2.00 s
Validation: passed

Confirmed missing inputs	x1 = 0 m; x2 = 10 m; v1 = +2 m/s; v2 = -3 m/s
Resolved end	catalog default time_limit at 10.00 s
Output	post-collision v1 = -4.00 m/s; v2 = +1.00 m/s
Simulation / run	2e7f400e-e3ca-4fc5-8ee6-8486365d97ff / 4e94f150-0c05-4c5c-b754-9d5df6811a9a

Validation result: PASS (all analytical checkpoints passed within tolerance 0.02).

Findings and test notes

The run exposed and corrected integration defects before the final pass: strict JSON schema required-field handling, ambiguity false positives around the unit m/s, legacy duplicate output keys, normalized angle roundoff, end-condition routing for collision prompts, and frontend architecture/scene-contract checks.

No unresolved ambiguity remained after the teacher-confirmed values were submitted. AI supplied extracted values and labels only; formulas and solver execution stayed in backend code.

Important limitation: the integrated browser session was unavailable in this environment. The three images embedded here are deterministic data renders from the validated `SimulationResponse` payloads, not direct DOM screenshots from a live browser canvas. Backend E2E and frontend static checks passed; a live browser smoke test should still be run when a browser session is available.

Artifacts

The three PNG snapshots are stored alongside this PDF in `output/pdf`. Raw validated responses used for the report are `linear-simulation.json`, `projectile-simulation.json`, and `collision-simulation.json` at the workspace root.